

2026 PSBH[®]

國際問題
解決松

03 — 06
SEPTEMBER

Problem Solving for Better Health[®]

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雙金榮耀・永續領航 | 2026 APSAA / TSAA 亞太暨臺灣永續行動獎雙金獎肯定
Double Gold Distinction | Honored with APSAA & TSAA Double Gold Sustainability Action Awards 2026



03我們如何透過智慧減壓防護、體位管理與術中監測，有效預防長時間手術所導致的術中壓傷並優化後續處置

03 How We Effectively Prevent Intraoperative Pressure Injuries Caused by Prolonged Surgeries and Optimize Subsequent Management Through Smart Pressure-Relief Protection, Positioning Management, and Intraoperative Monitoring

報告發表人 北醫 醫學二 鄭份庭 2026.09.06

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03 How We Effectively Prevent Intraoperative Pressure Injuries Caused by Prolonged Surgeries and Optimize Subsequent Management Through Smart Pressure-Relief Protection, Positioning Management, and Intraoperative Monitoring

北醫 醫學二 鄭份庭

Taipei Medical University, Medicine, Year 2, Yi-Ting Cheng

一、臨床痛點 Clinical Pain Points

術前 Preoperative

風險評估、擺位判斷缺乏個人化與客觀依據，仰賴護理師經驗
Subjective positioning; heavily reliant on nurse experience.

術中 Intraoperative

無術中監測，難持續觀察、掌握高風險部位的受壓狀況
Sterile draping prevents continuous monitoring of pressure on high-risk areas.

術後 Postoperative

缺乏客觀資料，造成追蹤、檢討與交班資訊的不足
Lack of data for tracking and handovers.

二、文獻探討 Literature Review

作者	論文名稱	出處	時間	方法結論
Shih et al.	Flexible Textile-Based Pressure Sensing System Applied in the Operating Room for Pressure Injury Monitoring of Cardiac Operation Patients	Sensors	2020	• 柔性紡織壓力感測墊 • 監測心臟手術患者術中介面壓力
Sving et al.	Using pressure mapping intraoperatively to prevent pressure ulcers—A quasi-experimental study	Health Science Reports	2023	• 116 名手術患者 • Pressure mapping 輔助評估微調體位
Yoo et al.	A wireless, skin-integrated system for continuous pressure distribution monitoring to prevent ulcers across various healthcare environments	npj Flexible Electronics	2026	• 皮膚整合式多點壓力感測 + BLE • 結合壓力與持續時間進行警示

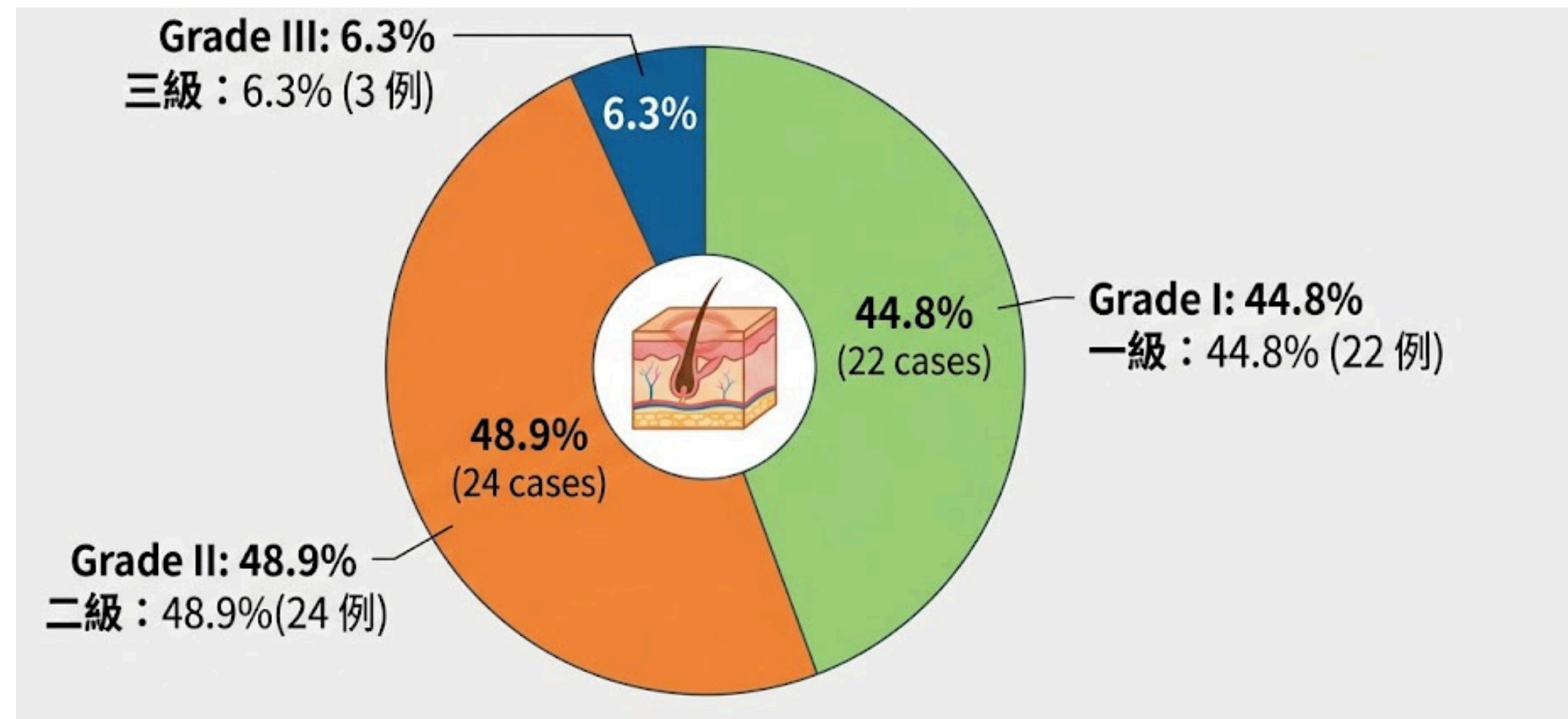
二、文獻探討-市場機會

Literature Review – Market Opportunity

- 手術後壓傷：顯著的經濟與臨床負擔 (Beriso et al., 2024)

Post-Operative Pressure Injury: A Significant Economic and Clinical Burden (Beriso et al., 2024)

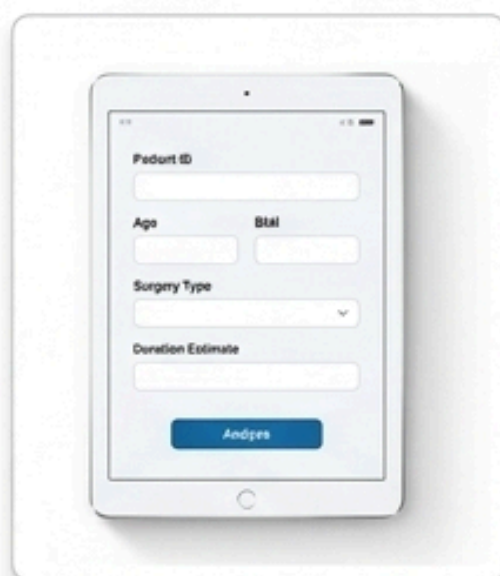
3~4倍
3–4 Times



三、解決方法-AI輔助術中壓傷智慧防護與動態風險管理系統

Solution – AI-Assisted Smart Intraoperative Pressure Injury Prevention and Dynamic Risk Management System

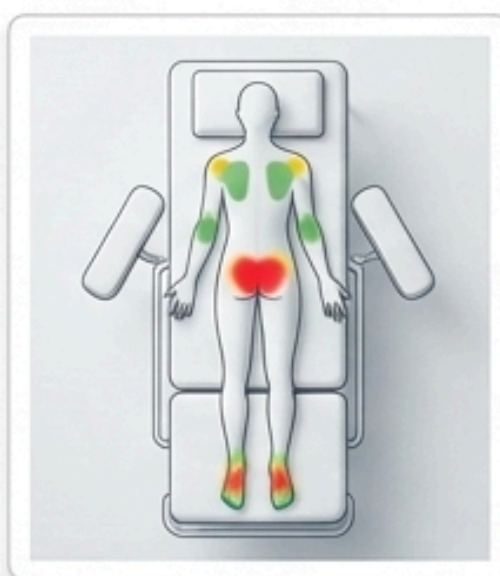
• 五大閉環機制 (5-Stage Closed-Loop Process)



系統功能
System
Action

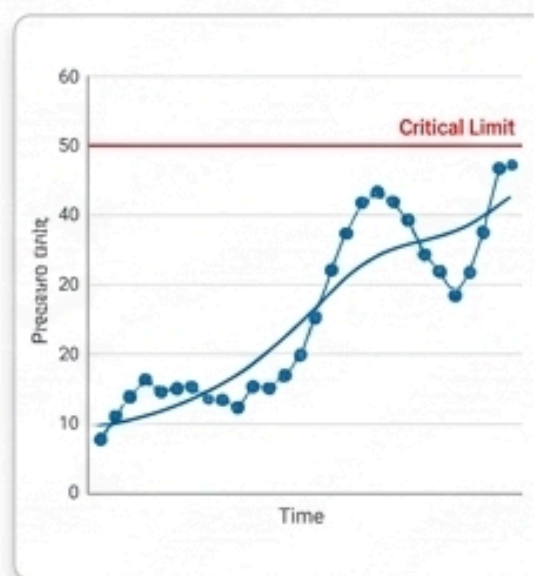
輸入病人條件與
手術數據

Input patient &
surgery data



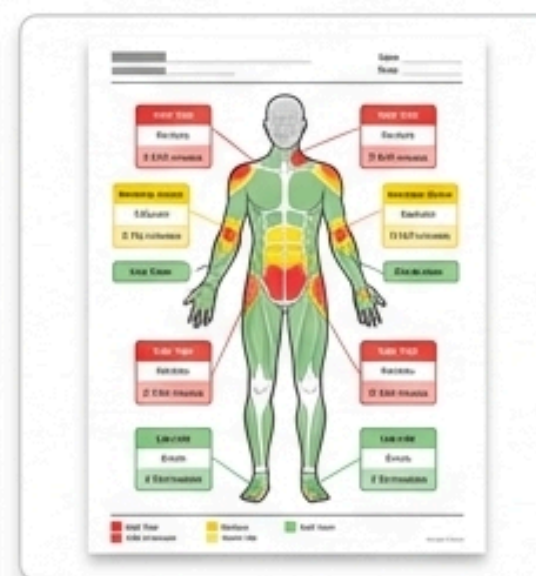
人體圖標示 (●●●)
高風險區

Body map indicates
high-risk areas



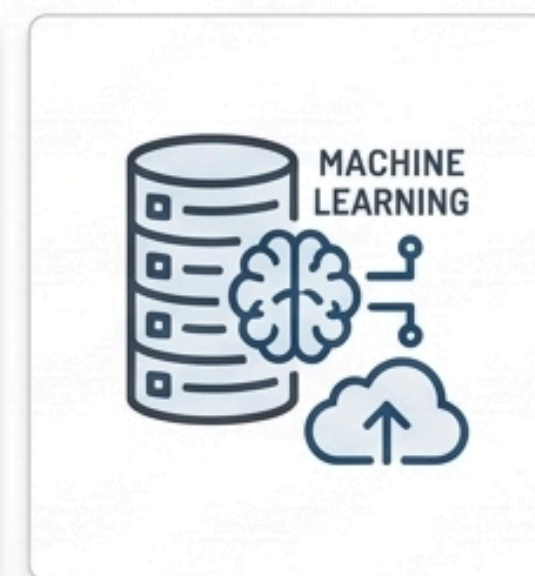
持續記錄壓力 × 時間
(Pressure × Duration)

Continuously log
Pressure × Duration



自動產生風險地圖
與累積時間

Auto-generate risk map
& cumulative time

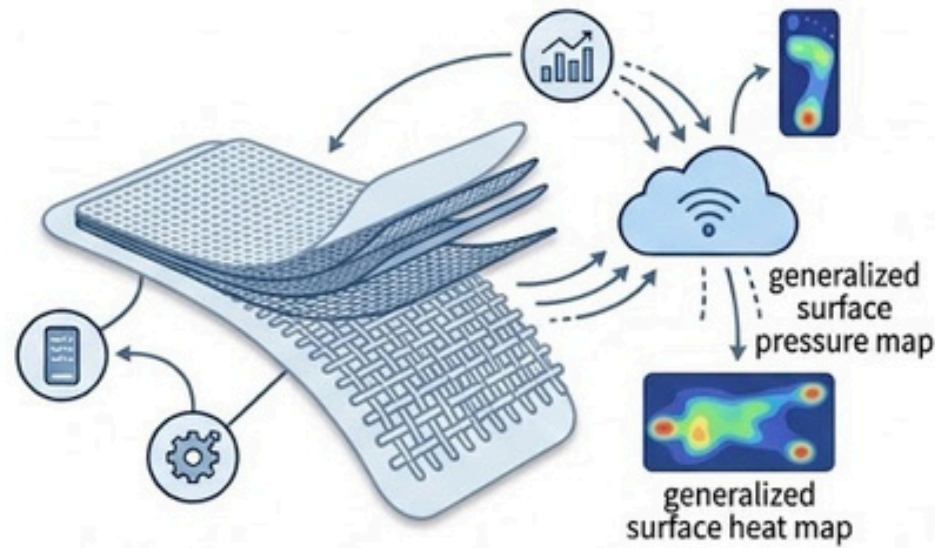


記錄最終結果
(Outcome)

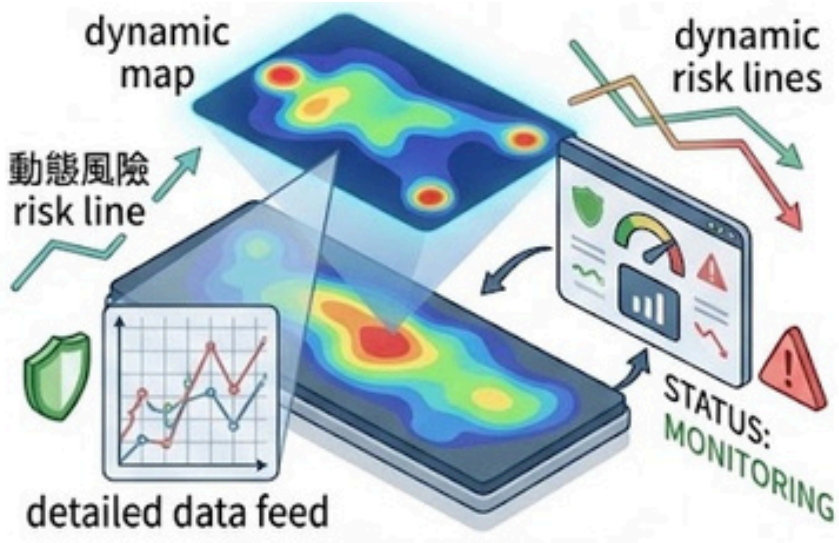
Record final outcome

① 超薄柔性低壓迫感測

Ultra-Thin Flexible Sensing



③動態風險評估 Dynamic Risk Assessment



②影像紀錄與責任歸屬之釐清

Image Documentation & Clear Accountability



④ 耗材與成本最佳化

Cost Efficiency & Risk Mitigation



五、競品分析 Competitive Analysis

比較項目 (Comparison Item)	Rogers 團隊系統 (Rogers Team System)	XSENSOR ForeSite OR	我們的產品 (Our Product)
感測架構 (Sensing Architecture)	單片固定形狀，點位固定 (Single fixed-shape unit, fixed point locations)	手術床整面感測墊，非貼身 (Full-surface sensing mat on OR table, not body-worn)	多片獨立貼片，擺位彈性 (Multiple independent patches, flexible positioning)
技術方法 (Technical Approach)	商用氣壓晶片 + BLE、無線充電 (Commercial pneumatic chip + BLE, wireless charging)	大面積感測床墊、高解析壓力圖 (Large-area sensing mattress, high-resolution pressure map)	開發中 (In development)
警示機制 (Alert Mechanism)	外接裝置震動 + LED警示 (External device vibration + LED alert)	壓力影像，無壓力 + 時間警示 (Pressure imaging, no pressure + time alert)	電腦紅黃綠燈顯示 (Computer red/yellow/green light display)
後續動作 (Follow-up Action)	僅提示翻身 (Only prompts repositioning)	可存資料研究，無個人化建議 (Data can be stored for research, no personalized advice)	分析後建議脂肪墊／敷料擺放 (Analyzes data to recommend fat pad/dressing placement)
應用場域 (Application Setting)	通用：居家／病房／ICU／OR (General use: home/ward/ICU/OR)	手術室專用，固定於手術床 (OR-specific, fixed to operating table)	聚焦手術室術前術中術後 (Focused on pre-, intra-, and post-op in OR)

六、營運模式 Business Model

基礎設施

Infrastructure

+

進階訂閱功能

Premium Add-ons (Subscription)

硬體 (Hardware) :

- 感測貼片 (Sensor Patch)

核心功能（Core Features）：

- 術中即時壓力監控、動態風險警示、術後交班卡及連通紀錄

(Real-time pressure monitoring, DRS alerts, and automated post-operative handoff records)

模組 A | AI 個人化擺位指引

術前智慧建議支撐與防護位置。

(Module A | AI-Guided Personalized Positioning: Smart preoperative recommendations for support and protection placement.)

模組 B | 跨科室品質改善與教學大數據

支援評鑑指標與護理訓練。

(Module B | Cross-Department Quality Improvement & Training Analytics: Supporting quality metrics and clinical training.)

七、團隊組成及能力(計畫可行性)Feasibility

1

臨床可行性

Clinical Feasibility

醫學系 Medicine

2

技術可行性

Technical Feasibility

化生 + 資工資科 Chem/Bio + CS/IS

3

執行可行性

Operational Feasibility

跨領域整合 Cross-Disciplinary

系所 Dept.	核心能力 Core Ability	關鍵貢獻 Key Contribution
● 醫學系 Medicine	臨床需求・病人安全 Clinical Needs & Patient Safety	對準真實痛點,不做無人用的技術 Target real pain points, avoid unused tech
● 化生系 Chem/Bio	感測材料・生物相容 Sensing Materials & Biocompatibility	穩定訊號感測,原型可驗證 Stable sensing, verifiable prototype
● 資工／資科系 CS/IS	系統開發・AI演算法 System Dev & AI Algorithms	即時數據處理與模型建置 Real-time data processing & modeling
● 數媒系 Digital Media	UI/UX・視覺溝通 UI/UX & Visual Communication	降低操作與學習門檻 Lower the learning curve

臨床 → 感測 → 演算 → 介面 完整閉環，跨領域即時反覆驗證

Clinical → Sensing → Algorithm → Interface — A Closed Loop, Verified Across Disciplines in Real Time

Thank You for Your Attention !

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